NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computing

Department of Software Engineering

QUALIFICATION: BACHELOR OF COMPUTER SCIENCE	
QUALIFICATION CODE: 07BACS	LEVEL: 7
COURSE: ARTIFICIAL INTELLIGENCE	COURSE CODE: ARI711S
SESSION: MAY 2023	PAPER: THEORY
DURATION: 2 HOURS	MARKS: 50

FIRST-OPPORTUNITY TEST	
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This paper consists of **6 pages** (including this front page)

INSTRUCTIONS

1. This paper contains 4 questions.
2. Answer all questions on the question paper.
3. Marks/scores are indicated at the right end of each question.
4. Calculators are permitted.
5. NUST examination rules and regulations apply.

NAME & SURNAME									
STUDENT NUMBER									

FOR STAFF USE ONLY	
QUESTION 1: SEARCH	
QUESTION 2: ADVERSARIAL SEARCH	
QUESTION 3: CSP	
QUESTION 4: MDP	
TOTAL MARK	
PERCENTAGE	

QUESTION 1: SEARCH **[10]**

Select whether the following statements are either TRUE or FALSE. Cross [X] the corresponding box for each question, to indicate your answer.

		True	False	
1.1	Greedy search can take longer to terminate than uniform-cost search.			(2)
1.2	Graph search can sometimes expand more nodes than tree search			(2)
1.3	Doubling your computer's speed allows you to double the depth of a tree search given the same amount of time.			(2)
1.4	Iterative deepening search is guaranteed to expand more nodes than breadth first search (on any graph whose root is not the goal).			(2)
1.5	A* search with a heuristic that is not completely admissible may still find the shortest path to the goal state.			(2)

QUESTION 2: ADVERSARIAL SEARCH **[10]**

Using the α – β pruning technique, solve the adversarial game depicted in Figure 1. Indicate a prune by drawing an [X] on the branch.

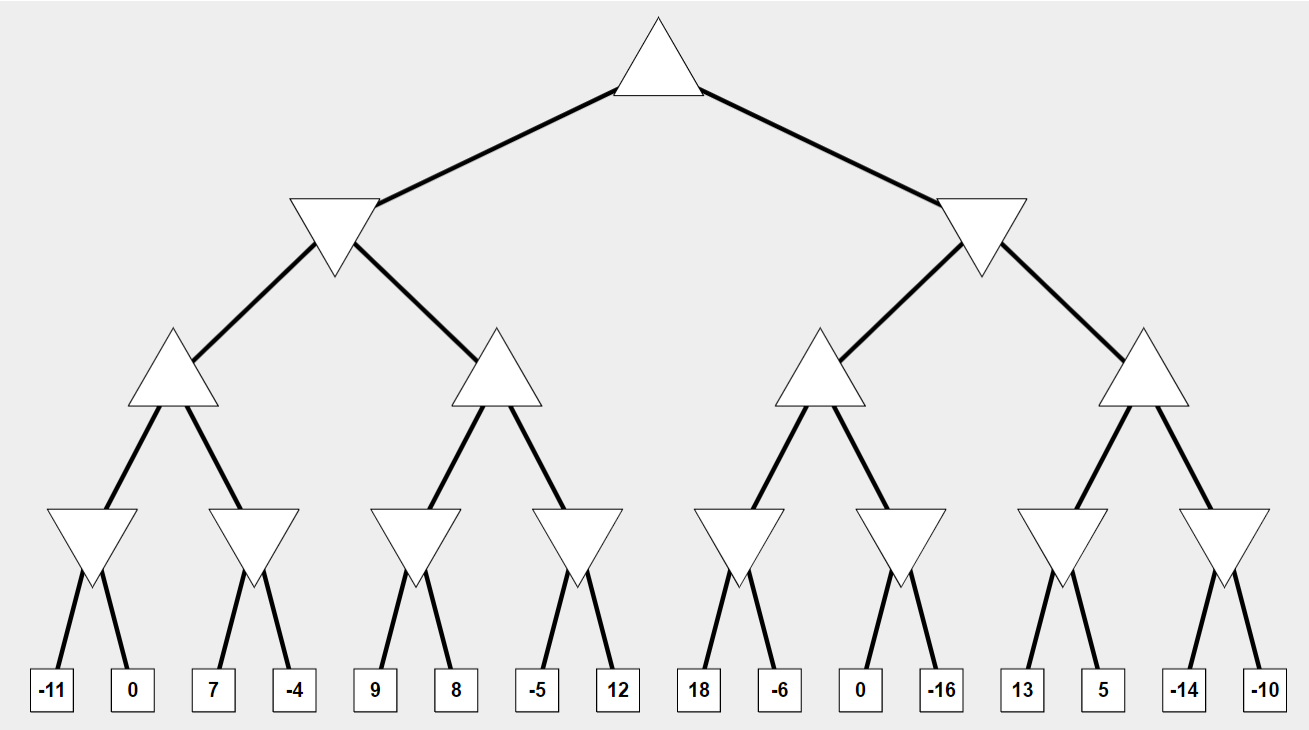

Figure 1: Adversarial Search Problem

QUESTION 3: CSP **[10]**

You are in charge of scheduling for computer science classes that meet Mondays, Wednesdays and Fridays. There are 5 classes that meet on these days and 3 professors who will be teaching these classes. You are constrained by the fact that each professor can only teach one class at a time.

The classes are:

- Class 1 - Intro to Programming: meets from 8:00-9:00am
- Class 2 - Intro to Artificial Intelligence: meets from 8:30-9:30am
- Class 3 - Natural Language Processing: meets from 9:00-10:00am
- Class 4 - Computer Vision: meets from 9:00-10:00am
- Class 5 - Machine Learning: meets from 10:30-11:30am

The professors are:

- Professor A, who is qualified to teach Classes 1, 2, and 5.
- Professor B, who is qualified to teach Classes 3, 4, and 5.
- Professor C, who is qualified to teach Classes 1, 3, and 4.

3.1 Formulate this problem as a CSP problem in which there is one variable per class, stating the domains, and constraints. Constraints should be specified formally and precisely, but may be implicit rather than explicit. (5)

3.2 Draw the constraint graph associated with your CSP. (1)

3.3 Provide a possible solution to the CSP scheduling problem. All variables should be assigned and no constraints broken. (4)

QUESTION 4: MDP **[20]**

Consider the blocks world. The blocks can be on a table or in a box. Consider three generic actions: a_0, a_1, and a_2 described as follows:
a_0: when applied to a block, will keep it in the box;
a_1: when applied to a block, will move it on the table;
a_2: when applied to two blocks, will move the first one on top of the second one.

Consider the following four states in the system:
S_0: all blocks are in the box, no block is on the table;
S_1: only block B is on the table; all other blocks are in the box;
S_2: both blocks B and C are on the table, with C on top of B;
S_3: blocks B, C and D are on the table, with D on top of C and C on top of B.

Furthermore, additional information is provided in Table 1, where each state has a reward, possible actions and a transition model for each action. Note that for a given action, the probability values indicated in its transition model all sum up to 1.

State	Reward	Action	Transition Model
S_0	r_0	a_{0b}	$(1, S_0)$
		a_{1b}	$(p_0, S_0); (p_1, S_1)$
S_1	r_1	a_{0c}	$(1, S_1)$
		a_{1c}	$(p_0^1, S_1); (p_1^1, S_4); (p_2^1, S_2)$
		a_{2c}	$(p_0^2, S_1); (p_1^2, S_2);$
S_2	r_2	a_{0d}	$(1, S_2)$
		a_{1d}	$(p_0^3, S_2); (p_1^3, S_5); (p_2^3, S_3)$
		a_{2d}	$(p_0^4, S_2); (p_1^4, S_3);$
S_3	100	-	-

Table 1: Additional Information

Assuming we model this problem as Markov Decision Process (MDP) and consider a discount value ϒ.

4.1 Provide the Bellman equation variation as applied to value iteration. (5)

4.2 Provide the utility of each of the states S_0, S_1 and S_2 for the first three iterations using the value iteration algorithm. Note that although the states S_4 and S_5 have not been defined, they should be assumed in the system. (10)

4.3 Consider the following policy, $\pi_0 = \{S_0 \rightarrow a_{0b}, S_1 \rightarrow a_{1c}, S_2 \rightarrow a_{2d}\}$. Is π_0 optimal? Explain. (5)

END OF TEST PAPER